

Numerics

Computed BN Angles and Evidence for the Divergence Condition

Diego Rincón

Independent

Abstract

We compute the Beurling–Nyman distances d_N^2 and alignment angles $\cos^2 \theta_N$ for $N = 1$ to 150, using the exact Gram matrix formula the exact Gram matrix from [1] and inner products $b_k = \log k/k$ from [2]. Key findings:

1. d_N^2 decreases from 1 to 0.009 by $N = 150$, consistent with $d_N \rightarrow 0$ (RH).
2. The partial sum $S_N = \sum_{j=1}^N \cos^2 \theta_j$ grows: $S_{10} \approx 2.63$, $S_{50} \approx 3.17$, $S_{150} \approx 3.41$.
3. The rate of growth is consistent with $S_N \sim C \log N / \log \log N$ ($C \approx 1.1$).
4. Primes dominate: $\cos^2 \theta_N$ is large when $k = N + 1$ is prime and small when k is composite. Primes account for $\sim 79\%$ of S_{50} .

The observed growth of S_N toward ∞ is strong numerical evidence for the angle divergence condition $\sum \cos^2 \theta_N = \infty$, which is equivalent to the Riemann Hypothesis.

1 The Computation

Definition 1 (Exact Gram matrix). For $k, l \geq 2$ with $m = \text{lcm}(k, l)$:

$$G_{kl} = \frac{1}{klm} \sum_{r=1}^{m-1} (r \bmod k)(r \bmod l) \left(\psi\left(\frac{r+1}{m}\right) - \psi\left(\frac{r}{m}\right) \right),$$

where $\psi = \Gamma'/\Gamma$ is the digamma function. The $r = 0$ term vanishes since $(0 \bmod k)(0 \bmod l) = 0$. The formula is exact (no approximation) and derived in [1].

Definition 2 (Algorithm). Given N :

1. Build $G_N \in \mathbb{R}^{N \times N}$ with $(G_N)_{ij} = G_{i+1, j+1}$ for $i, j \in \{1, \dots, N\}$ (using $k = i + 1$, $l = j + 1$).
2. Build $b \in \mathbb{R}^N$ with $b_i = \log(i + 1)/(i + 1)$.
3. Solve $G_N x = b$ (Cholesky: G_N is positive definite).
4. $d_N^2 = 1 - b^\top x$.
5. $\cos^2 \theta_N = 1 - d_N^2/d_{N-1}^2$.

The Gram matrix G_N is symmetric positive definite by construction.

2 The Data

Table 1 gives d_N^2 , $\cos^2 \theta_N$, and the partial sum $S_N = \sum_{j=1}^N \cos^2 \theta_j$ for $N = 1$ to 50. An asterisk marks when $k = N + 1$ is prime.

N	k	prime	d_N^2	$\cos^2 \theta_N$	S_N
1	2	*	0.30685	0.69315	0.693
2	3	*	0.09575	0.68797	1.381
3	4		0.06595	0.31119	1.692
4	5	*	0.03632	0.44931	2.142
5	6		0.03621	0.00306	2.145
6	7	*	0.02439	0.32644	2.471
7	8		0.02425	0.00590	2.477
8	9		0.02410	0.00616	2.483
9	10		0.02385	0.01007	2.493
10	11	*	0.02050	0.14057	2.634
11	12		0.02050	0.00007	2.634
12	13	*	0.01987	0.03044	2.664
13	14		0.01853	0.06755	2.732
14	15		0.01838	0.00818	2.740
15	16		0.01794	0.02415	2.764
16	17	*	0.01711	0.04625	2.810
17	18		0.01710	0.00059	2.811
18	19	*	0.01685	0.01424	2.825
19	20		0.01654	0.01868	2.844
20	21		0.01612	0.02504	2.869

N	k	prime	d_N^2	$\cos^2 \theta_N$	S_N
21	22		0.01609	0.00207	2.871
22	23	*	0.01558	0.03186	2.903
23	24		0.01558	0.00002	2.903
24	25		0.01555	0.00173	2.905
25	26		0.01545	0.00623	2.911
26	27		0.01534	0.00712	2.918
27	28		0.01524	0.00650	2.924
28	29	*	0.01473	0.03371	2.958
29	30		0.01456	0.01178	2.970
30	31	*	0.01456	0.00004	2.970
31	32		0.01408	0.03292	3.003
32	33		0.01352	0.03942	3.042
33	34		0.01330	0.01635	3.059
34	35		0.01327	0.00211	3.061
35	36		0.01318	0.00724	3.068
36	37	*	0.01309	0.00640	3.074
37	38		0.01299	0.00819	3.083
38	39		0.01299	0.00001	3.083
39	40		0.01274	0.01921	3.102
40	41	*	0.01235	0.03063	3.133

50	51		0.01185		3.173
100	101	*	0.01015		3.328
150	151	*	0.00939		3.405

Table 1: BN distances d_N^2 , angles $\cos^2 \theta_N$, and partial sums S_N . Asterisks mark $k = N + 1$ prime. Initial values: $d_0^2 = 1$, $\cos^2 \theta_1 = \log 2$ (exact).

3 Observations

Observation 3 (Monotone decrease of d_N^2). d_N^2 decreases from 1 (no projection) to 0.00939 at $N = 150$. The rate of decrease slows but shows no sign of leveling off. Extrapolating: $d_N^2 \approx A/(\log N)^\alpha$ for some $A, \alpha > 0$. A fit to the data $N \in [10, 150]$ gives $\alpha \approx 1.5$: $d_N^2 \approx 0.30/(\log N)^{1.5}$. Under this fit: $d_{10^6}^2 \approx 0.30/(6 \log 10)^{1.5} \approx 0.001$, still positive — but the trend is toward 0.

Observation 4 (Prime dominance). When $k = N + 1$ is prime, $\cos^2 \theta_N$ is substantially larger than at composite k . For $N \leq 50$: the 15 primes ($k = 2, 3, 5, \dots, 47$) account for 2.495 of the total $S_{50} = 3.173$, i.e., 78.6% of the alignment sum. Average per-prime contribution: 0.166. Average per-composite contribution: 0.019. The prime-to-composite ratio is ≈ 8.7 .

The prime at $k = 2$ gives $\cos^2 \theta_1 = \log 2 \approx 0.693$ (exact), the dominant single contribution. The primes $k = 3, 5, 7$ each contribute ≈ 0.3 – 0.7 . The primes $k = 11, 13, \dots$ contribute 0.03 – 0.14 .

Explanation: $\rho_{1/p}$ (for prime p) is nearly orthogonal to $\mathcal{V}_{p-2} = \text{span}\{\rho_{1/2}, \dots, \rho_{1/(p-1)}\}$, because primes introduce new arithmetic structure not captured by smaller k . Composites $k = ab$ overlap with $\rho_{1/a}$ and $\rho_{1/b}$ already present in the span, reducing their orthogonal component v_k .

Observation 5 (Rate of growth of S_N). The ratio $S_N/\log N$ is: 1.14 ($N = 10$), 0.96 ($N = 20$), 0.81 ($N = 50$), 0.72 ($N = 100$), 0.68 ($N = 150$). The decreasing ratio rules out $S_N \sim C \log N$.

The ratio $S_N \cdot \log \log N / \log N$: 0.96 ($N = 10$), 1.05 ($N = 20$), 1.11 ($N = 50$), 1.10 ($N = 100$), 1.09 ($N = 150$). This ratio appears to be stabilizing near 1.1, suggesting:

$$S_N \approx 1.1 \cdot \frac{\log N}{\log \log N}.$$

Since $\log N / \log \log N \rightarrow \infty$, this is consistent with $S_N \rightarrow \infty$, which is the angle divergence condition ($\sum \cos^2 \theta_j = \infty$, i.e., RH).

Remark 6 (Connection to Mertens' theorem). Observation 4 and 5 together suggest:

$$S_N \approx C_1 \sum_{p \leq N+1} \cos^2 \theta_{p-1} + C_2 \sum_{k \leq N, k \text{ composite}} \cos^2 \theta_{k-1}.$$

If the prime contributions satisfy $\cos^2 \theta_{p-1} \sim A/\log p$, then $\sum_{p \leq N} \cos^2 \theta_{p-1} \sim A\pi(N) \sim AN/\log N$. If instead $\cos^2 \theta_{p-1} \sim A/p$, then $\sum_{p \leq N} \cos^2 \theta_{p-1} \sim A \log \log N$.

But the data shows $S_N \sim C \log N / \log \log N$, faster than $\log \log N$. This implies either the composite contributions are significant at large N , or the prime contributions decay slower than $1/p$. The data at $N = 150$ with $S_N = 3.41$ and $\log \log 150 \approx 1.61$ gives $S_N / \log \log N \approx 2.11$, still much larger than 1.

The precise rate of S_N remains to be determined analytically.

4 Evidence for RH

Theorem 7 (Numerical evidence for RH). The computational data for $N = 1$ to 150 is consistent with:

1. $d_N^2 \rightarrow 0$ (the Beurling–Nyman criterion for RH): the distance decreases from 1 to 0.009 and shows no sign of stabilizing.
2. $\sum \cos^2 \theta_N = \infty$ (the angle criterion for RH [3]): S_N appears to grow as $\log N / \log \log N$, which diverges.
3. The prime structure of the BN basis determines the rate: primes are the “non-redundant” directions; composites add minor corrections.

This is consistent with, but does not prove, the Riemann Hypothesis. A proof would require showing either $d_N \rightarrow 0$ or $S_N \rightarrow \infty$ rigorously. The present computation provides strong evidence and identifies the prime structure as the key mechanism.

Remark 8 (The rate $\log N / \log \log N$). The growth rate $S_N \sim C \log N / \log \log N$ is interesting from two perspectives:

1. It is faster than $\log \log N$ (so Mertens-type bounds don't explain it).
2. It is slower than $\log N$ (so naive estimates would miss it).
3. The factor $1 / \log \log N$ suggests logarithmic corrections from the interaction between the prime and composite contributions.

Whether the rate is exactly $\log N / \log \log N$ requires larger N or an analytic argument.

Remark 9 (What larger N would show). Under $S_N \sim 1.1 \log N / \log \log N$: $S_{10^6} \approx 5.8$ and $S_{10^{20}} \approx 13.2$. The sum grows slowly but steadily. Computing to $N = 10^6$ requires a fast Gram algorithm: $O(N^2)$ with the Levinson recursion (see [1]).

References

- [1] Rincón, D. (2026). The Gram Matrix. Preprint.
- [2] Rincón, D. (2026). The Approximation. Preprint.
- [3] Rincón, D. (2026). The Angle: The Riemann Hypothesis as a Divergence Condition. Preprint.
- [4] Rincón, D. (2026). Entropy: The Beurling–Nyman Flow as Lyapunov Function for RH. Preprint.
- [5] Rincón, D. (2026). Oscillatory: The Inner Product as an Oscillatory Integral. Preprint.
- [6] Báez-Duarte, L. (2005). A strengthening of the Nyman–Beurling criterion. *Atti della Accademia Nazionale dei Lincei* 14(1), 5–11.
- [7] Odlyzko, A.M. (1992). The 10^{20} -th zero of the Riemann zeta function and 175 million of its neighbors. Unpublished manuscript.